

Notes on Lewin Ch4: Spectral Theorem and Functional Calculus

1 Multiplication Operators

Let $B \subset \mathbb{R}^d$ be Borel, and let μ be a locally finite Borel measure on B . We set $\mathfrak{H} = L^2(B, d\mu; \mathbb{C})$. Local finiteness implies that $L_c^\infty \subset \mathfrak{H}$.

Example 1. Let $B = \{x_1, \dots, x_k\} \subset \mathbb{R}^d$ and $\mu = \sum_i \delta_{x_i}$. Then $L^2(B, d\mu) \cong \mathbb{C}^k$, so every operator T identifies under the obvious choice of basis $e_i = \mathbb{1}_{\{x_i\}}$ with a matrix $M \in \mathbb{C}^{k \times k}$, with $M_{ij} = \langle e_i, T e_j \rangle = (T e_j)(x_i)$. Then $g = T f$ is the element $g(x_i) = \sum_j M_{i,j} f(x_j)$.

- A diagonal matrix M corresponds to $f \mapsto a f$ where $a(x_i) = M_{ii}$.
- Every Hermitian matrix $M = M^*$ can be diagonalized (possibly under a different basis).
- Let $a \in L_{\text{loc}}^2(B, d\mu)$, then M_a is the operator defined by

$$M_a f(x) = a(x) f(x), \quad D(M_a) = \{v \in L^2(B, d\mu) : av \in L^2(B, d\mu)\}. \quad (1)$$

Theorem 1. Let $a \in L_{\text{loc}}^2(B, d\mu)$.

1. $(M_a, D(M_a))$ is closed.
2. $\sigma(M_a) = \text{ess ran}(a)$, where the essential range of a is

$$\text{ess ran}(a) = \{y \in \mathbb{C} : \mu(|a(\cdot) - y| \leq \varepsilon) > 0 \text{ for all } \varepsilon > 0\}. \quad (2)$$

1. The eigenvalues of M_a are the $\lambda \in \text{ess ran}(a)$ such that $\mu(a = \lambda) > 0$, with the corresponding eigenspace $L^2(\{a = \lambda\}, d\mu)$, the space of all square-integrable functions with support in the set $\{a = \lambda\}$, defined μ -a.e.
 2. $(M_a, D(M_a))$ is bounded iff $a \in L^\infty(B, d\mu)$.
 3. $(M_a, D(M_a))$ is self-adjoint iff a is real-valued (bounded or not).
- Bits of the proof. For $\lambda \notin \text{ess ran}(a)$, there is $\varepsilon > 0$ such that $|a - \lambda| > \varepsilon$ μ -a.e. Thus, $\frac{1}{a - \lambda} \in L^\infty(B, d\mu)$, and the map $v \mapsto v(a - \lambda)^{-1}$ is bounded $L^2(B, d\mu) \rightarrow D(M_a)$ (using $\frac{a}{a - \lambda} = 1 + \frac{\lambda}{a - \lambda}$) and is an inverse for $M_a - \lambda$. For $\lambda \in \text{ess ran}(a)$, there exists $R_n > 0$ such that $\mu(\{|a - y| < \frac{1}{n}\} \cap \mathbb{B}_{R_n}) \in (0, \infty)$ for all n . Then with $u_n := \mathbb{1}_{\{|a - y| < \frac{1}{n}\} \cap \mathbb{B}_{R_n}}$ we have

$$\|(M_a - \lambda)u_n\|^2 \leq \frac{1}{n^2} \|u_n\|^2. \quad (3)$$

So $M_a - \lambda$ cannot be invertible.

Theorem 2 (Theorem 4.4: Spectral Theorem). Let $(A, D(A))$ be self-adjoint on $\mathfrak{H}$. Then there exists $d \geq 1$, a Borel set B , a locally finite measure μ on B , a real-valued locally bounded function $a \in L_{\text{loc}}^\infty(B, d\mu)$, and an isomorphism $U : \mathfrak{H} \rightarrow L^2(B, d\mu)$ such that

$$U A U^{-1} = M_a, \quad U D(A) = D(M_a). \quad (4)$$

One can take $d = 2$, $B = \sigma(A) \times \mathbb{N} \subset \mathbb{R}^2$, $a(s, n) = s$, and μ a finite measure on B .

Corollary (resolvent bound).

$$\|(A - z)^{-1}\| = \frac{1}{d(z, \sigma(A))}. \quad (5)$$

Corollary (isolated eigenvalues). Every isolated point of the spectrum is an eigenvalue.

- Define $f(A) := U^{-1}M_{f(a)}U$, $D(f(A)) = U^{-1}D(M_{f(a)})$ for any such isomorphism. We need to show this is independent of the choice of U .

Theorem 3 (Theorem 4.8: Functional Calculus for bounded Borel functions). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$f \in \mathcal{L}^\infty(\mathbb{R}, \mathbb{C}) \mapsto f(A) \in \mathcal{B}(\mathfrak{H}) \quad (6)$$

defined on the C^* -algebra $\mathcal{L}^\infty(\mathbb{R}, \mathbb{C})$ of bounded Borel functions on $\mathbb{R}$, with values in the algebra $\mathcal{B}(\mathfrak{H})$ of bounded operators on $\mathfrak{H}$, such that:

- (i) it is a morphism of C^* -algebras ($\mathbb{C}$ -linear, preserves product and star operation).
- (ii) it is continuous, with $\|f(A)\| \leq \sup_{x \in \mathbb{R}} |f(x)|$.
- (iii) if $f(x) = (x - z)^{-1}$ with $z \in \mathbb{C} \setminus \mathbb{R}$, then $f(A) = (A - z)^{-1}$.
- (iv) if $f|_{\sigma(A)} \equiv 0$ then $f(A) = 0$.
- (v) if $|f_n(x)| \leq C$ and $f_n \rightarrow f$ pointwise on $\mathbb{R}$, then $f_{n(A)}v \rightarrow f(A)v$ for all $v \in \mathfrak{H}$.

Remark 1 (Remark 4.9: Spectral measure). Let v be a unit vector of H . By Theorem 3, the map

$$f \in C_{b(\mathbb{R}, \mathbb{R})}^0 \mapsto \varphi_{v(f)} := \langle v, f(A)v \rangle \in \mathbb{R} \quad (7)$$

is a continuous linear form. If in addition $f \geq 0$ we can write

$$f(A) = \sqrt{f(A)}^2, \quad (8)$$

which shows that φ_v is a positive linear form on C_b^0 . Hence by Riesz–Markov, there is a unique Borel probability measure $\mu_{A,v}$ on $\mathbb{R}$ such that

$$\langle v, f(A)v \rangle = \int_{\mathbb{R}} f(s) d\mu_{A,v}(s). \quad (9)$$

With $(B, \mu) = (\sigma(A) \times \mathbb{N}, \mu)$ and $a(s, n) = s$ from Theorem 2, if $U : H \rightarrow L^2(B, \mu)$ is the corresponding unitary map, we can write

$$\langle v, f(A)v \rangle = \langle Uv, f(a)Uv \rangle_{L^2(B, \mu)} = \int_B f(a) |Uv|^2 d\mu = \int_{\sigma(A) \times \mathbb{N}} f(s) |Uv(s, n)|^2 d\mu(s, n). \quad (10)$$

Therefore $\mu_{A,v}$ is the pushforward measure¹

$$\mu_{A,v} = a_*(|Uv|^2 \mu), \quad (12)$$

i.e. for every Borel set $E \subset \mathbb{R}$,

$$\mu_{A,v}(E) = \int_{a^{-1}(E)} |Uv(x)|^2 d\mu(x) = \int_{E \times \mathbb{N}} |Uv(s, n)|^2 d\mu(s, n). \quad (13)$$

In words, $\mu_{A,v}$ is the cylindrical projection on $\sigma(A)$ of the probability measure $|Uv(s, n)|^2 d\mu(s, n)$ on $\sigma(A) \times \mathbb{N}$. We then have $v \in D(A)$ iff $\mu_{A,v}$ has a moment of order two, and in this case

$$\int_{\mathbb{R}} s^2 d\mu_{A,v} = \|Av\|^2. \quad (14)$$

We also have

$$\int_{\mathbb{R}} s d\mu_{A,v}(s) = \langle v, Av \rangle. \quad (15)$$

We will in fact need this construction in the proof of Theorem 2.

For every bounded Borel function f , $f(A)$ is defined by Theorem 3. By setting $E_{A(B)} := \mathbb{1}_{B(A)}$ we obtain the spectral theorem in terms of projection-valued measures:

¹The book writes

$$d\mu_{A,v}(s) = \sum_{n \in \mathbb{N}} |Uv(s, n)|^2 d\mu(s, n) \quad (11)$$

but this is a little fast and loose with notation; the sum is a partial integration of $d\mu(s, n)$ that comes from the pushforward. The proper way is to define the slice measures $\mu_{n(E)} = \mu(E \times \{n\})$, then we can write $d\mu_{A,v}(s) = \sum_{n \in \mathbb{N}} |Uv(s, n)|^2 d\mu_{n(s)}$.

Theorem 4 (Projection-valued spectral measure). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$E_A : \mathcal{B}(\mathbb{R}) \rightarrow \mathcal{B}(H) \quad (16)$$

from the Borel subsets of $\mathbb{R}$ to the orthogonal projections on H , such that:

(i) $E_{A(\emptyset)} = 0$ and $E_{A(\mathbb{R})} = I$.

(ii) if $(B_n)_n$ are pairwise disjoint Borel subsets of $\mathbb{R}$, then

$$E_{A(\cup_n B_n)} v = \sum_n E_{A(B_n)} v \quad (17)$$

for every $v \in H$, where the series converges in H .

(iii) for every pair of Borel subsets $B, C \subset \mathbb{R}$,

$$E_{A(B \cap C)} = E_{A(B)} E_{A(C)}. \quad (18)$$

(iv) for every bounded Borel function $f \in \mathcal{L}^\infty(\mathbb{R}, \mathbb{C})$,

$$f(A) = \int_{\mathbb{R}} f(s) dE_{A(s)}. \quad (19)$$

(v) We can recover

$$A = \int_{\mathbb{R}} s dE_{A(s)}. \quad (20)$$

Definition 1 (Scalar spectral measure). Let v be a unit vector of H . Then $\mu_{A,v}$ and E_A are related by

$$\mu_{A,v}(B) = \langle v, E_{A(B)} v \rangle. \quad (21)$$

Corollary (Corollary 4.10: Functional Calculus for locally bounded Borel functions). Let $(A, D(A))$ be self-adjoint and let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a locally bounded Borel function. Then $f(A)$ defined above is independent of the isomorphism U used to represent A as a multiplication operator.

This follows from the functional calculus for bounded functions because we can describe $D(f(A))$ and $f(A)v$ in terms of the corresponding functional calculus for $f_n = f \mathbb{1}_{\{|f| < n\}}$,

$$\begin{aligned} v \in D(f(A)) &\iff \limsup_{n \rightarrow \infty} \|f_n(A)v\| < \infty, \\ f(A)v &= \lim_{n \rightarrow \infty} f_n(A)v. \end{aligned} \quad (22)$$

2 Proof of Theorems 4.4 and 4.8

• Structure of the proof:

1. First we prove the functional calculus for the “resolvent algebra” $\mathcal{A}$ of resolvents. This gives Theorem 5 using the Stone–Weierstrass theorem. This calculus is for functions in $C_{\text{lim}}^0 := \mathbb{C} + C_0^0(\mathbb{R}, \mathbb{C})$ (continuous functions with equal limits at $\pm\infty$).
2. Deduce Theorem 2.
3. Use the monotone class theorem to deduce uniqueness and hence Theorem 3 (the other properties of Theorem 3 follow directly from the definition).

• C_{lim}^0 is a unital C^* -algebra.

Theorem 5 (Theorem 4.11: Continuous functional calculus). Let $(A, D(A))$ be self-adjoint. There exists a unique map

$$f \in C_{\lim(\mathbb{R}, \mathbb{C})}^0 \mapsto f(A) \in \mathcal{B}(\mathfrak{H}) \quad (23)$$

such that:

- (i) it is a morphism of C^* -algebras ($\mathbb{C}$ -linear, preserves product and star operation).
- (ii) it is continuous, with $\|f(A)\| \leq \sup_{x \in \mathbb{R}} |f(x)|$.
- (iii) if $f(x) = (x - z)^{-1}$ with $z \in \mathbb{C} \setminus \mathbb{R}$, then $f(A) = (A - z)^{-1}$.

Proof. Define the resolvent algebra $\mathcal{A}$ as the algebra generated by constant functions and the rational maps $x \mapsto (x - z)^{-1}$ for $z \in \mathbb{C} \setminus \mathbb{R}$. By the Stone–Weierstrass theorem, $\mathcal{A}$ is dense in $C_{\lim}^0$. It is clearly a C^* -algebra.

We first show that there is a unique morphism of C^* -algebras $\mathcal{A} \rightarrow \mathcal{B}(\mathfrak{H})$ sending $(x - z)^{-1}$ to $(A - z)^{-1}$. We are forced to directly map $(x - z)^{-1}$ to $(A - z)^{-1}$ by (iii) and similarly for linear combinations of products by (i). That this assignment is unique and well-defined follows from

- $((A - z)^{-1})^* = (A - \bar{z})^{-1}$, and
- the resolvent identity

$$(A - z)^{-1} - (A - w)^{-1} = (w - z)(A - z)^{-1}(A - w)^{-1}, \quad (24)$$

which demonstrates commutativity.

For (ii) we will use Lemma 1.

Lemma 1 gives us (ii) by using

$$\|f\|_{\infty}^2 - |f(x)|^2 \in \mathcal{A}. \quad (25)$$

This continuity allows us to well-define $f(A)$ for all $f \in C_{\lim}^0$ by approximating by elements of $\mathcal{A}$ (via Stone–Weierstrass) and using the continuity to show that the limit is independent of the choice of approximating sequence. Uniqueness follows from the density of $\mathcal{A}$ in $C_{\lim}^0$. ■

Lemma 1 (Lemma: Stability under the square root). Let $f \in \mathcal{A}$ be non-negative on $\mathbb{R}$. Then there is a unique $g \in \mathcal{A}$ such that $g^2 = f$ and $g \geq 0$ on $\mathbb{R}$. In particular,

$$f(A) = g(A)^2 \geq 0. \quad (26)$$

Proof. Let us write bold letters $\alpha, p, q, \dots$ to denote vectors / multiindices of some length and set $(f(\alpha))^\beta$ to mean $\prod_{i(f(\alpha_i))}^{\beta_i}$ when α and β are the same length.

Note that every $f \in \mathcal{A}$ can be written as a reduced rational function $f = \frac{P}{Q}$ where P and Q are polynomials with no common roots, and Q has no real roots. Conversely every such rational function is in $\mathcal{A}$ by decomposing as partial fractions.

Let for $f = \frac{P}{Q} \in \mathcal{A}$,

$$P = c(x - \alpha)^p (x - z)^{p'}, \quad Q = (x - \xi)^q, \quad (27)$$

with $\alpha_i \in \mathbb{R}$, $z_i \in \mathbb{C}$, $\xi_i \in \mathbb{C}$.

Next suppose that $f \geq 0$ on $\mathbb{R}$. In particular then $f(x) \in \mathbb{R}$ when $x \in \mathbb{R}$, so

$$P\overline{Q} = Q\overline{P}, \quad (28)$$

which implies that $c \in \mathbb{R}$ by comparing the leading term, and also that

$$(x - z)^{p'} (x - \bar{\xi})^q = (x - \xi)^q (x - \bar{z})^{p'}. \quad (29)$$

Since the fraction $\frac{P}{Q}$ is reduced, none of the z_i can equal the ξ_i , so each $(x - z_i)^{p'_i}$ must be a factor of $(x - \bar{z})^q$. Similarly, each $(x - \bar{\xi}_i)^{q_i}$ must be a factor of $(x - \xi)^q$. It follows that the complex numbers appear with their conjugates with equal multiplicity, so

$$f(x) = c(x - \alpha)^p \frac{|x - z|^{2p'}}{|x - \xi|^{2q}}. \quad (30)$$

Now, using that $f \geq 0$ on $\mathbb{R}$, we have $c \geq 0$ and p_i are even. So we can take

$$g(x) = \sqrt{c}(x - \alpha)^{\frac{p}{2}} \frac{|x - z|^{p'}}{|x - \xi|^q}, \quad (31)$$

as required. ■

Proof (of the Spectral Theorem, Theorem 2). As in Remark 1, we can construct the scalar spectral measure $\mu_{A,v}$ for some unit vector v , as follows. By the Riesz–Markov representation theorem, there is a unique Borel probability measure $\mu_{A,v}$ on $\mathbb{R}$ such that

$$\langle v, f(A)v \rangle = \int_{\mathbb{R}} f(s) d\mu_{A,v}(s) \quad (32)$$

for every bounded continuous function f . Now observe that

$$\langle g(A)v, f(A)v \rangle = \int_{\mathbb{R}} \overline{g(s)} f(s) d\mu_{A,v}(s) \quad (33)$$

so that the map $f \mapsto f(A)v$ is an isometry $C_{\lim(\mathbb{R},\mathbb{C})}^0 \rightarrow \mathfrak{H}$ with respect to the inner product. By taking the closure, this gives an isometry

$$U : L^2(\mathbb{R}, d\mu_{A,v}) \rightarrow \mathcal{X}_v := \overline{\{f(A)v : f \in C_{\lim(\mathbb{R},\mathbb{C})}^0\}}, \quad f \mapsto f(A)v. \quad (34)$$

Note that

$$\langle g(A)v, (A-z)^{-1}f(A)v \rangle = \langle v, (\overline{g}(\bullet - z)^{-1}f)(A)v \rangle = \frac{\int_{\mathbb{R}} \overline{g(s)} f(s) d\mu_{A,v}(s)}{s-z} \quad (35)$$

Thus, extending by continuity, we see that $(A-z)^{-1}$ restricted to $\mathcal{X}_v$ is unitarily equivalent to multiplication by $(s-z)^{-1}$. It follows that A is unitarily equivalent to multiplication by s . Indeed, let

$$B := UAU^{-1} \quad (36)$$

Then $(B-z)^{-1} = U(A-z)^{-1}U^{-1} = M_{(s-z)^{-1}}$. At the same time, from

$$(s-z)(M_s - z)^{-1}f = (M_s - z)(M_s - z)^{-1}f = f \quad (37)$$

we see that $(M_s - z)^{-1} = M_{(s-z)^{-1}}$ as well. In particular the resolvents' ranges are the same, so $D(B) = D(M_s)$, and $I = (B-z)(M_s - z)^{-1}$ gives that $B = M_s$.

If there exists v such that $\mathcal{X}_v = \mathfrak{H}$, then we are done. If not, we need to iterate the argument.

Lemma 2 (Invariance of $\mathcal{X}_v$). The subspace $\mathcal{X}_v$ is invariant under $(A-z)^{-1}$, and $\mathcal{X}_v^\perp$ is also invariant under $(A-z)^{-1}$, for every $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof (of lemma). $(A-z)^{-1}\mathcal{X}_v \subset \mathcal{X}_v$ is because $(A-z)^{-1}f \in C_{\lim(\mathbb{R},\mathbb{C})}^0$ for every $f \in C_{\lim(\mathbb{R},\mathbb{C})}^0$, and $(A-z)^{-1}$ is continuous.

And for a similar reason, $(A-z)^{-1}\mathcal{X}_v^\perp \subset \mathcal{X}_v^\perp$, as $\langle f(A)v, (A-z)^{-1}w \rangle = \langle (A-\bar{z})^{-1}f(A)v, w \rangle = \langle ((\bullet - \bar{z})^{-1}f)(A)v, w \rangle = 0$. ■

Now, we can write

$$\mathfrak{H} = \bigoplus_{n \in \mathbb{N}} \mathcal{X}_{v_n} \quad (38)$$

as follows - let e_n be an orthonormal basis for $\mathfrak{H}$, and put $v_1 = e_1$. Then, let $v_2 = P_{\mathcal{X}_{v_1}}^\perp e_j$ (orthogonal projection) where e_j is the first basis vector not in $\mathcal{X}_{v_1}$. Note that $\mathcal{X}_{v_2} \perp \mathcal{X}_{v_1}$, since $v_2 \perp \mathcal{X}_{v_1}$, and

$$\langle f(A)v_1, g(A)v_2 \rangle = \langle v_1, (\overline{f}g)(A)v_2 \rangle = 0. \quad (39)$$

Continuing in this way gives the desired decomposition of $\mathfrak{H}$ into a direct sum of invariant subspaces.

We now have that $(A-z)^{-1}$ on each invariant subspace $\mathcal{X}_{v_n}$ is unitarily equivalent to multiplication by $(s-z)^{-1}$ on $L^2(\mathbb{R}, d\mu_{A,v_n})$. We can combine them into a single isomorphism to $L^2(B, d\mu)$ with $B = \mathbb{R} \times \mathbb{N}$ and $\mu(V \times \{n\}) = 2^{-n}\mu_{A,v_n}(V)$, and define $a(s, n) = s$. It follows as before that A is unitarily equivalent to multiplication by s . This completes the proof of Theorem 2. TODO $\mu_{A,v}$ vanishes outside of $\sigma(A)$ ■